



Crop Production and Management

By Sanskriti

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INTRODUCTION

What is Agriculture?

- Production of crops on a large scale is called agriculture
- Horticulture is the art of growing garden plants (fruits, vegetables, flowers)
- Agriculture is still the largest occupation in our country
- Nearly 70% of the population depends on agriculture

CROP CLASSIFICATION BY SEASONS

Crop Seasons (Based on Time of Cultivation)

Rabi Season

Time: October to March

Crops: Wheat, barley, pea, gram, potato, mustard

Harvesting: March to April

Kharif Season

Time: June to September

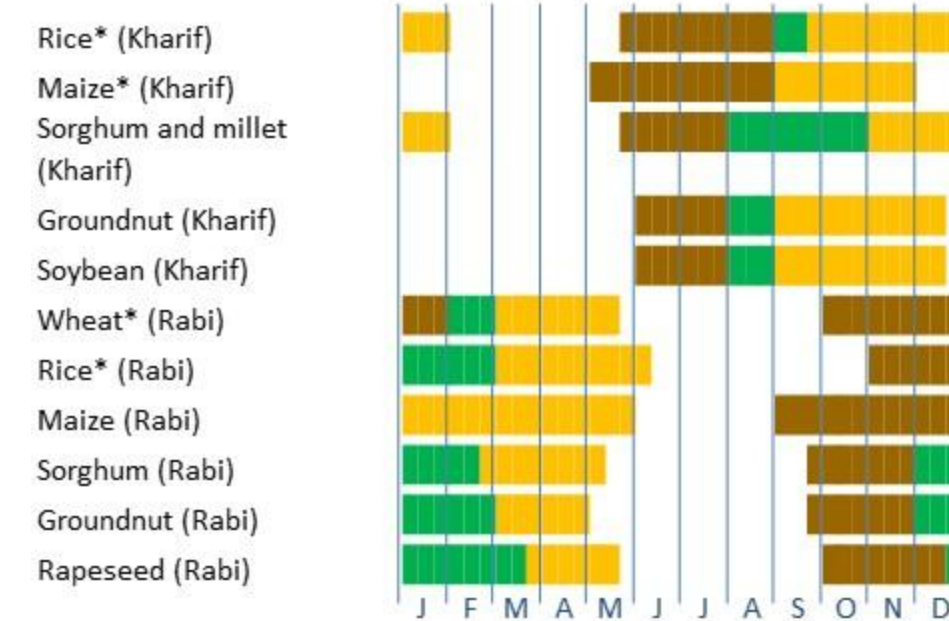
Crops: Rice, maize, jowar, bajra, tur, moong, urad, cotton, jute, groundnut, sugarcane

Harvesting: September to October

India

Crop Calendar

(*major foodcrop)



Sowing
Growing
Harvesting

CROP CLASSIFICATION BY PURPOSE

Crop Group	Examples
Cereals	Wheat, rice, maize, jowar, bajra, barley
Pulses	Gram, tur, urad, moong, masoor, khesari
Vegetables	Tomato, potato, cabbage, onion, cauliflower
Fruits	Apple, plum, peaches, grapes, guava
Oil seeds	Mustard, linseed, sunflower, castor
Fibre	Cotton, jute, hemp, flax
Sugar yielding crops	Sugarcane, sugar beet
Medicinal crops	Belladonna, mint, tulsi, ashgol, cinchona
Spice crops	Turmeric, coriander, cumin, chilli

BASIC AGRICULTURAL PRACTICES

Essential Steps in Crop Production

- 1. Preparation of soil**
- 2. Sowing**
- 3. Adding manures and fertilizers**
- 4. Irrigation**
- 5. Protection from weeds**
- 6. Harvesting**
- 7. Storage**

SOIL PREPARATION

Key Steps:

1. Ploughing and Digging

Loosening and turning up of soil

Done using wooden plough or iron leveller

Benefits:

- Allows roots to penetrate deep

- Brings nutrients to surface

- Buries weeds and previous crop remains

2. Levelling

Breaking big soil lumps called crumbs

Done with help of wooden plank or iron leveller

Creates uniform field surface

3. Manuring

Adding manure before tilling

Helps in proper mixing of manure with soil



AGRICULTURAL IMPLEMENTS

Traditional Tools:

- **Plough:** For breaking and loosening soil
- **Hoe:** For removing weeds
- **Cultivator:** For breaking soil after ploughing
- **Seed drill:** For sowing seeds

Modern Equipment:

- **Tractor-driven plough**
- **Cultivator**
- **Seed drill machine**
- **Harrow**
- **Leveller**

SOWING

Methods of Sowing:

1. Broadcasting

Seeds scattered manually by directly casting them in the soil
Simple but not very efficient

2. Seed Drill Method

Seeds sown manually at correct depth using furrows
More precise and efficient
Prevents seed wastage

Key Points:

Seed rate: Right amount of seeds per unit area

Depth: Proper depth for different crops

Spacing: Adequate distance between seeds

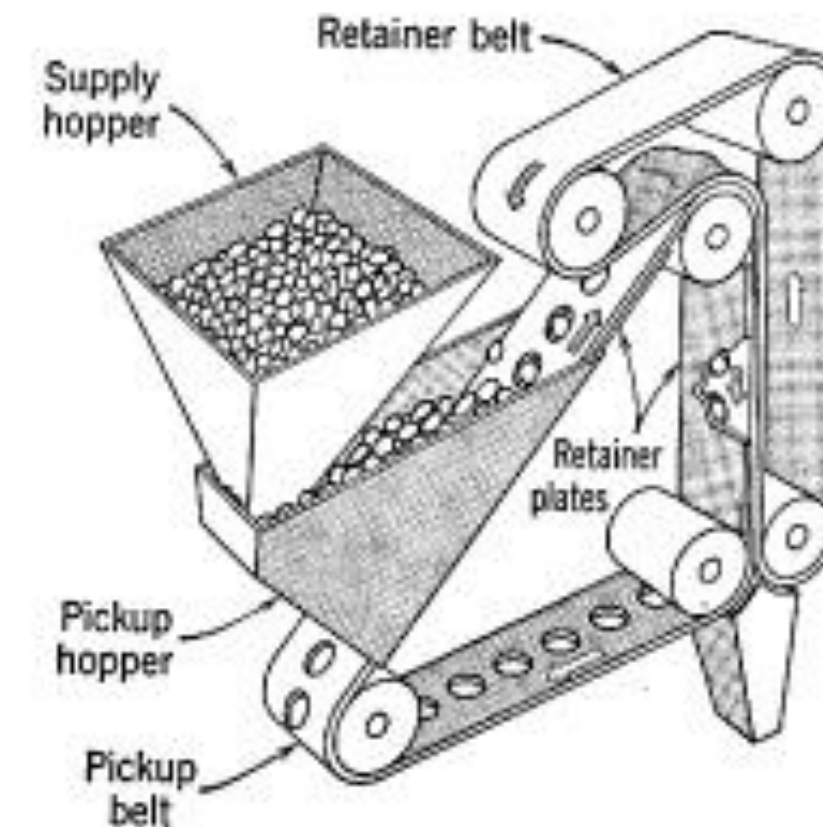


Fig. 11.6. Experimental belt-type planter developed for high-speed drilling of seeds such as peanuts. (J. G. Futral and R. L. Allen. *Agr. Eng.*, April, 1951.)

TRANSPLANTATION

What is Transplantation?

Seeds first germinated in nurseries

Seedlings transferred to main field

Process: Known as transplantation

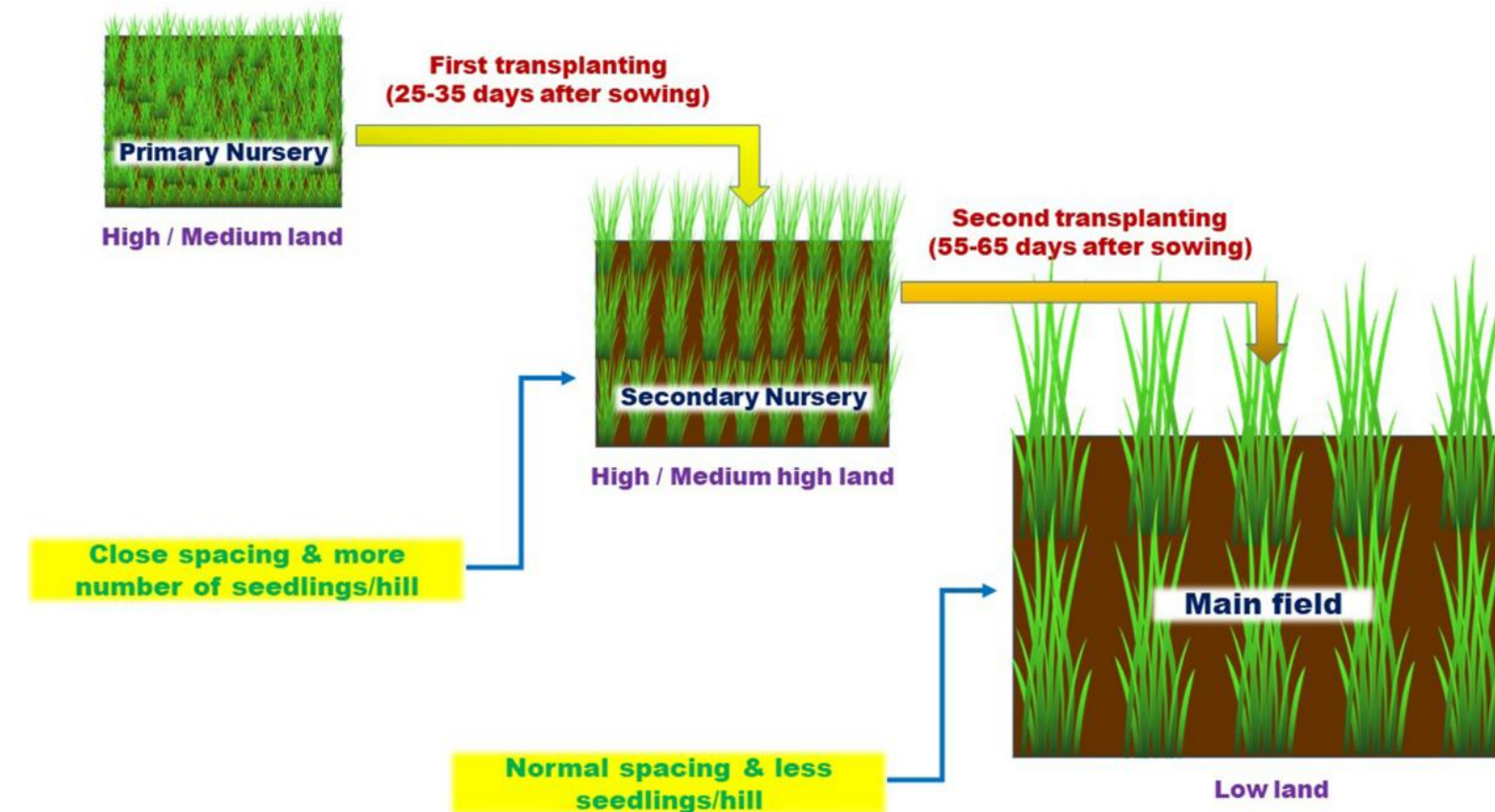
Crops Using Transplantation:

Rice

Tomato

Tobacco

Onion



MANURES AND FERTILIZERS

Manures (Natural)

Definition: Natural organic substances from decomposition of animal wastes and plant residues

Types:

Farm Yard Manure: Cattle dung, urine, straw, leaves

Compost: Decomposition of dead plants and animal wastes

Green Manure: Fresh plant materials buried in soil

Fertilizers (Chemical)

NPK: Nitrogen, Phosphorus, Potassium

Characteristics:

Nutrient specific

Quick acting

Expensive

Can harm soil if overused

	Manures	Fertilisers
(i)	They are not easily soluble in water. So they are not easily absorbed by the roots of the plants.	They are soluble in water and are absorbed by the plants easily.
(ii)	These are inorganic substances.	These are organic substances.
(iii)	Excessive use does not harm the soil texture, they provide humus to the soil.	Excessive use can change the chemical composition of the soil and also pollutes water.
(iv)	They restore the soil texture and help in water retention.	They may damage the soil texture.
(v)	These are nutrient specific.	These are not nutrient specific.

IRRIGATION

Sources of Water:

Rivers, reservoirs, lakes, ponds, wells, tube-wells, dams, canals

Traditional Methods:

Moat (Pulley system)

Chain pump

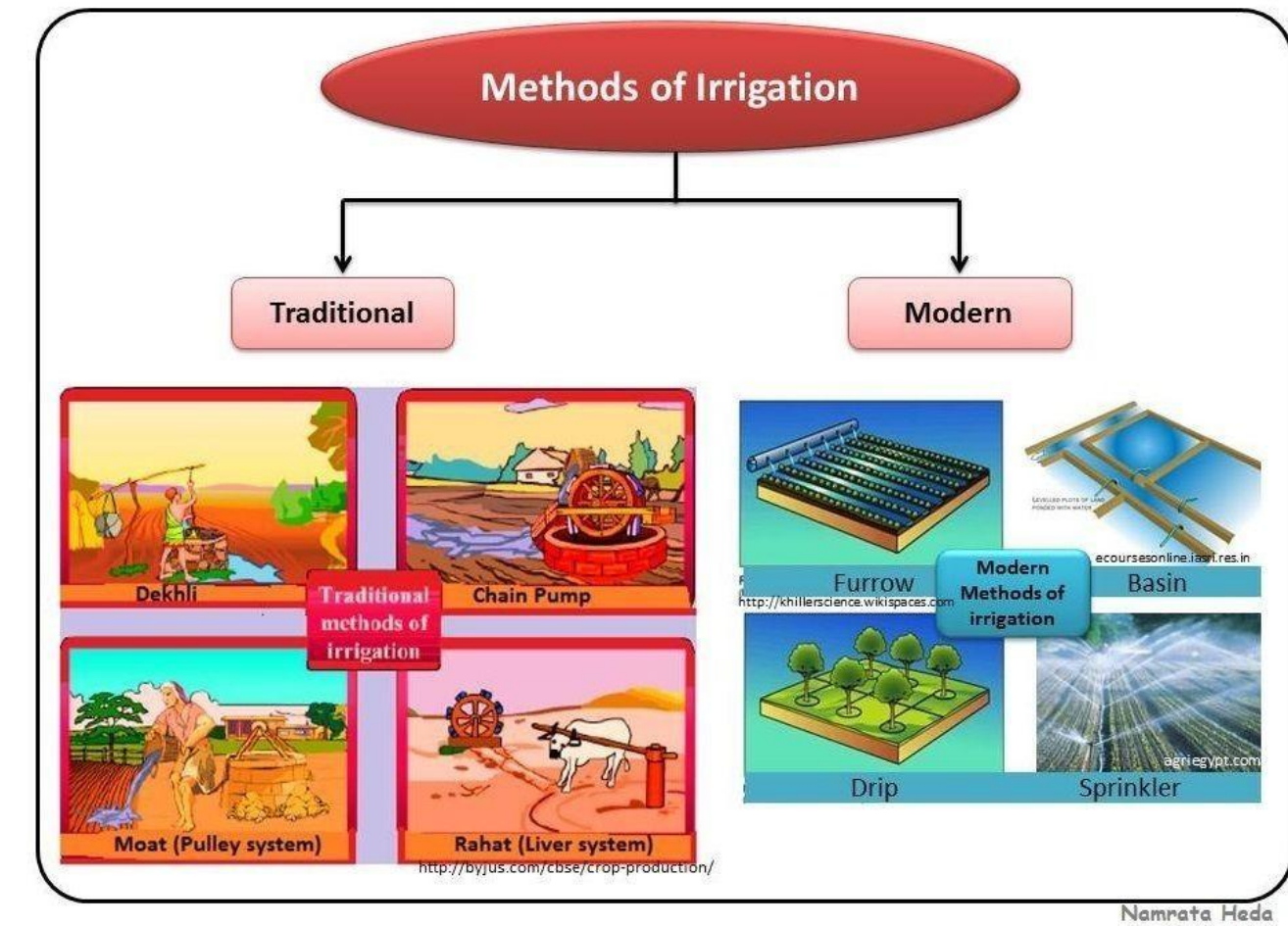
Dhekli

Rahat

Modern Methods:

Sprinkler irrigation: Water sprinkled through sprinklers

Drip irrigation: Water applied drop by drop to plant roots



IRRIGATION ADVANTAGES & DISADVANTAGES

Advantages:

Provides water at different stages of plant development

Maintains soil moisture for proper growth

Disadvantages of Excessive Irrigation:

Water logging: Roots don't get sufficient oxygen

Soil salinity: Accumulation of salts in soil

Strong winds: Can damage waterlogged plants

WEEDING

What are Weeds?

Unwanted plants growing alongside main crops

Examples: Wild oat, grass, Amaranthus, Chenopodium, Parthenium

Problems Caused by Weeds:

Compete for water, nutrients, space, light

Reduce crop yield

Some produce poisonous substances

Control Methods:

Manual weeding: Hand removal

Mechanical weeding: Using tools

Chemical weeding: Weedicides/herbicides

Chemical Method



Physical Method



Cultural Method



Biological Method



Dandelion



Sow thistle



Henbit



Shepherd's
purse



Pennywort



Annual
fleabane

CROP PROTECTION

Pests and Diseases:

Types of Pests:

Insects: Aphids, stem borers

Rodents: Rats, mice

Birds: Damage grains

Diseases: Fungi, bacteria, viruses

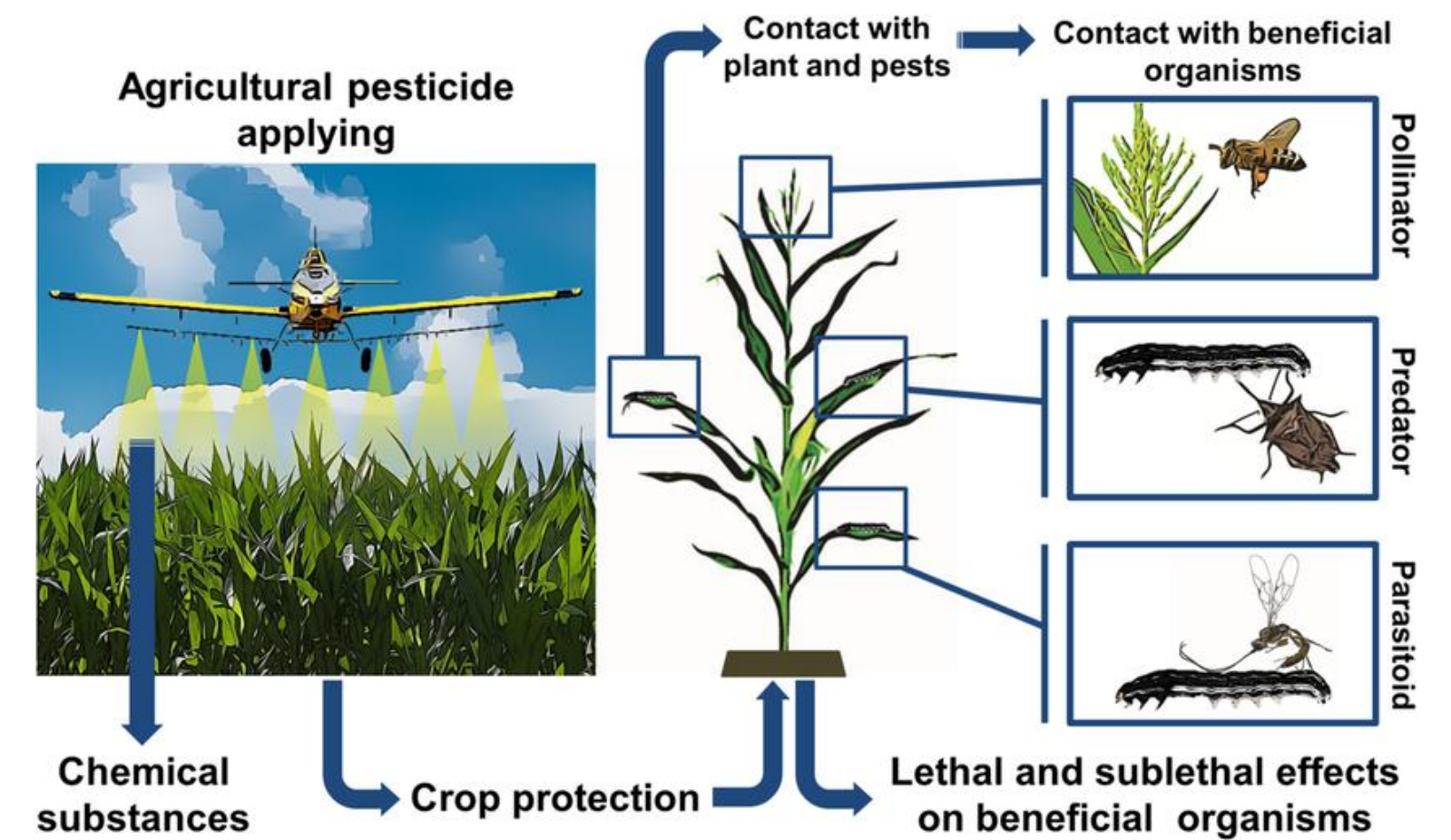
Control Methods:

Pesticides: Kill insects and pests

Insecticides: Specifically for insects (DDT, BHC, Malathion)

Fungicides: Control fungal diseases (Sulfur, lime-sulfur)

Rodenticides: Control rodents (Zinc phosphide, Warfarin)



HARVESTING

Definition:

Process of cutting and gathering matured crop

Methods:

Manual Harvesting:

Using sickle for cutting

Labor intensive but selective

Machine Harvesting:

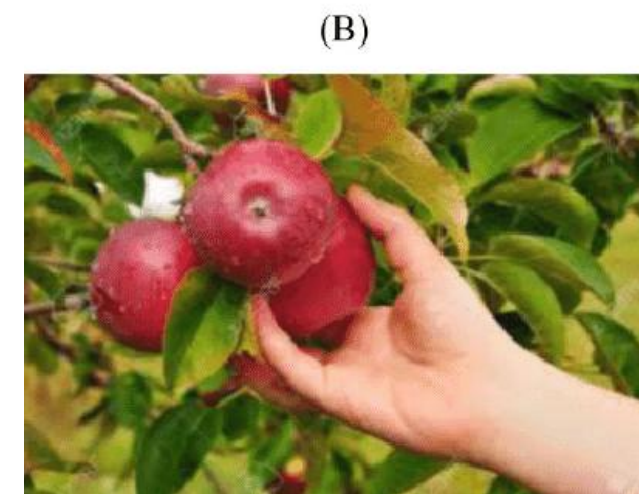
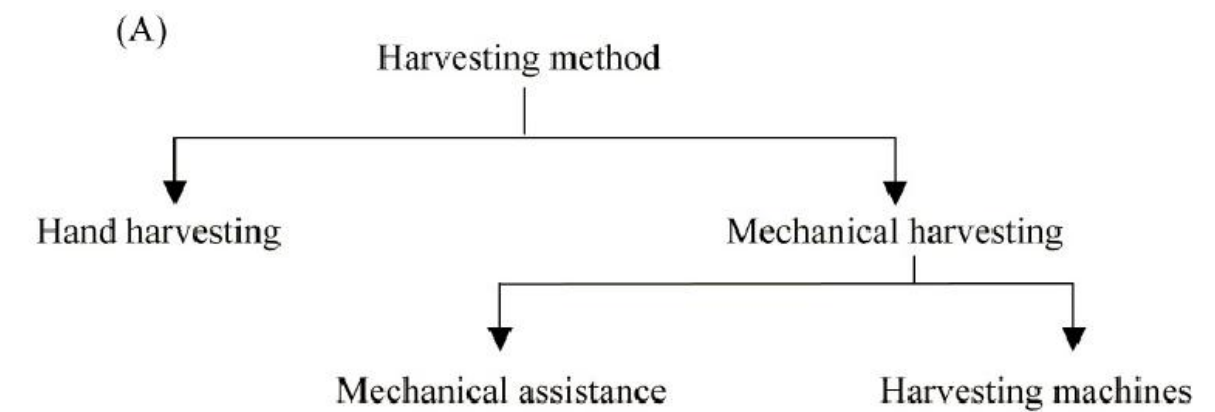
Using harvesters and combines

Faster and efficient for large areas

Grain Processing:

Threshing: Separating grain from harvested stalks

Winnowing: Separating grain from chaff using wind



STORAGE

Need for Storage:

Prevent spoilage by microorganisms

Protection from rodents and insects

Maintain market supply throughout year

Types of Storage:

Perishable Materials:

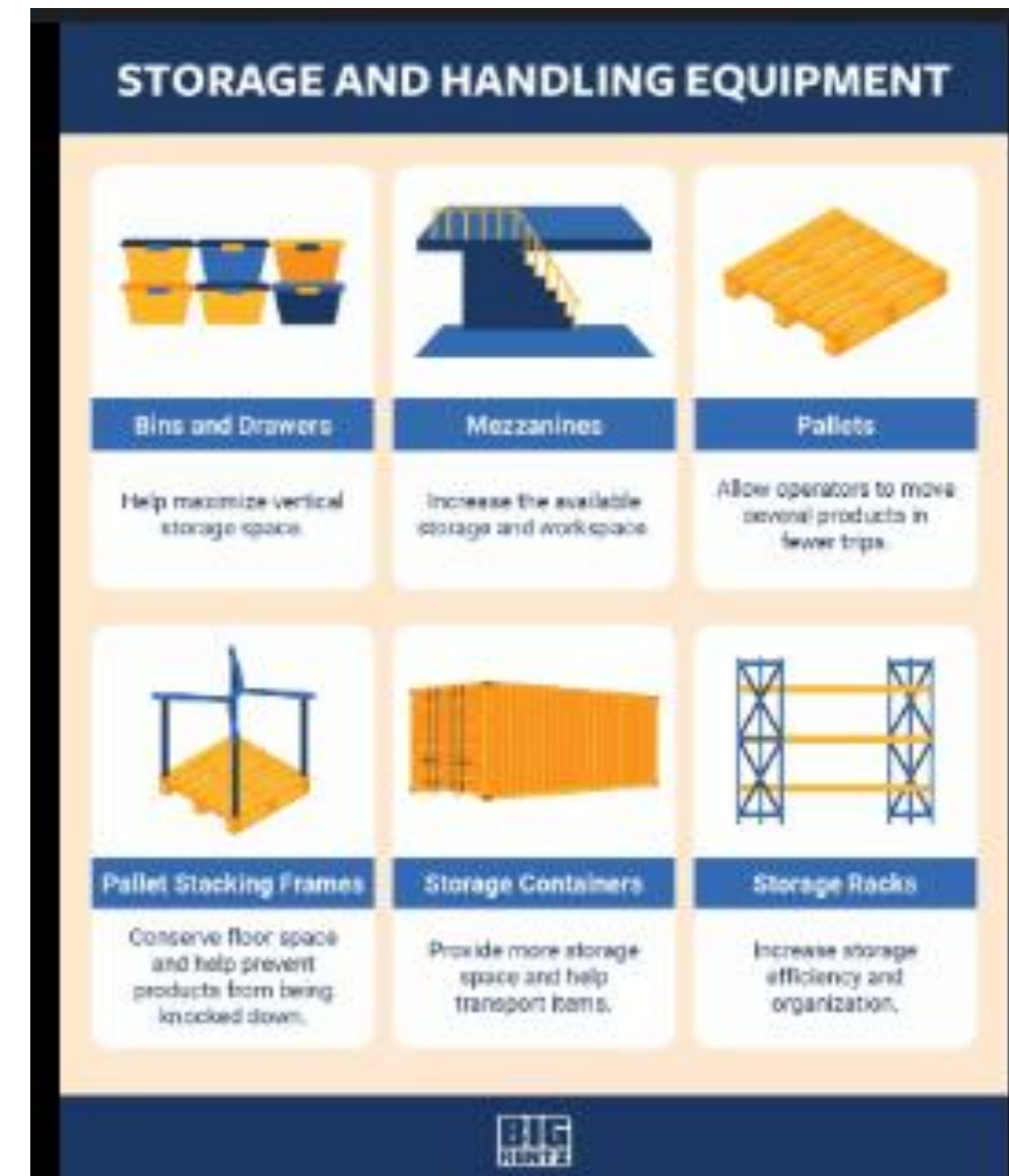
Cold storage: For vegetables, fruits, fish, meat, milk

Refrigeration: Low temperature preservation

Non-perishable Materials:

Dry storage: For grains, pulses at room temperature

Silos and granaries: Large scale storage



FOOD FROM ANIMALS

Animal Husbandry:

Definition: Management, breeding, feeding of domestic animals

Important Animals:

Dairy animals: Cows, buffaloes (milk production)

Poultry: Chickens, ducks (eggs and meat)

Fish farming: Aquaculture for fish production

Bee keeping: Apiculture for honey production

CATTLE BREEDS

Milk Producing Breeds:

Foreign breeds: Jersey, Brown Swiss, Holstein-Friesian

Indian breeds: Gir, Sahiwal, Red Sindhi, Deoni

Characteristics:

High milk yield

Disease resistance

Adaptation to local climate



Yakanaji cow (White Fulani)



Bokoloji bull (Sokoto Gudali)



Bodeeji/Bororo cow (Red Fulani)



Djelliji bull (Peulh nigérien)



Somba cow



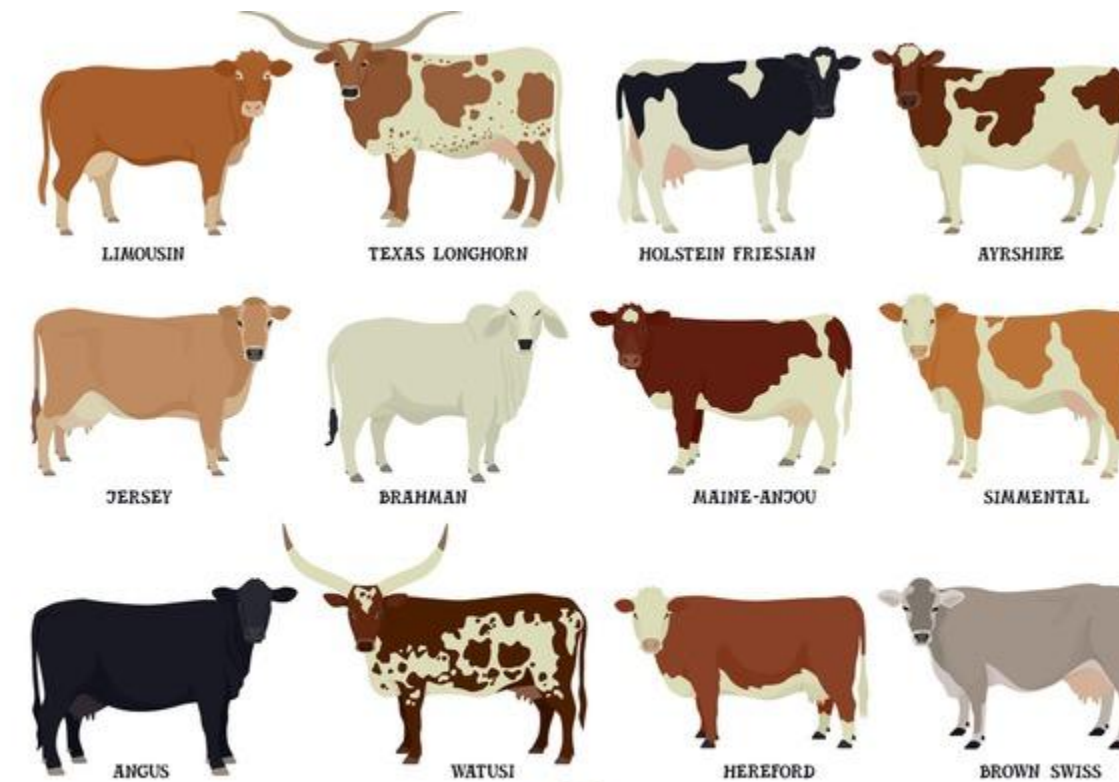
Boboji cow (Lagune cattle)



Bargouji/Muti cow (Borgou)



Dageeji cow / Dage



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CATTLE BREEDS

POULTRY AND FISH FARMING

Poultry:

Layers: For egg production

Broilers: For meat production

Common breeds: Rhode Island Red, White Leghorn, Aseel

Fish Farming:

Marine water fish: Pomfret, tuna, mackerel

Fresh water fish: Rohu, Catla, Mrigals

Composite fish culture: Multiple species together



CATLA



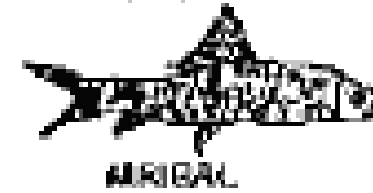
SILVER CARP



ROHU



GRASS CARP



MRIGAL



COMMON CARP

SUMMARY & KEY POINTS

Important Concepts for Olympiad:

- **Crop seasons and their characteristics**
- **Agricultural practices sequence**
- **Difference between manures and fertilizers**
- **Traditional vs modern farming methods**
- **Irrigation techniques and water management**
- **Pest control and crop protection**
- **Storage methods for different crops**
- **Animal husbandry practices**

